FITBUDDY

Smart Fitness & Nutrition Assistant

Project Documentation

Project Title: FitBuddy – Smart Fitness & Nutrition Assistant

Purpose: A simple web-based application that helps users generate structured workout plans and receive practical nutrition suggestions.

1. Introduction

FitBuddy is a fitness-assistance application designed to provide structured workout guidance and practical nutrition suggestions. It combines a web interface with backend services so users can interact with fitness-related features in one place.

2. Objectives

• Provide personalized-style workout plans based on user inputs.
• Give quick nutrition suggestions.
• Present information through a simple and user-friendly interface.
• Organize fitness data and requests through a backend API.

3. Key Features

• 7-day workout plan generation and updating
• Practical nutrition tips
• User-friendly web interface
• API-based backend
• Database-ready architecture
• Environment-variable based configuration for API keys

4. Technologies Used

Frontend: HTML, CSS, JavaScript
Backend: Python, FastAPI, Uvicorn
Database layer: SQLAlchemy
AI integration: Google Generative AI SDK
Development: Python virtual environment

5. System Architecture

User → Web Interface → FastAPI Backend → Fitness/AI Services → Response → Web Interface. The backend receives requests from the frontend, processes the request, optionally calls an AI service, and returns structured results.

6. Workout Module

The workout module is intended for structured 7-day workout generation and plan updating. User requirements can be passed to the backend, which prepares a workout response that can be displayed in the application.

7. Nutrition Module

The nutrition module provides fast, practical suggestions related to healthy eating and nutrition. It is designed for lightweight requests and easy integration with the main application.

8. Installation

Create a virtual environment, activate it, and install the dependencies from requirements.txt. Configure required environment variables such as API credentials before starting the backend server.

9. Sample Run

Start the FastAPI application with Uvicorn and open the frontend in a browser. The user can then enter fitness-related information and request a workout plan or nutrition suggestion.

10. Future Enhancements

Possible improvements include user accounts, progress tracking, exercise videos, reminders, dashboard charts, wearable-device integration, and stronger personalization.

11. Conclusion

FitBuddy provides a compact platform for fitness planning and nutrition guidance. Its modular design makes it suitable for further development as a larger fitness-support application.